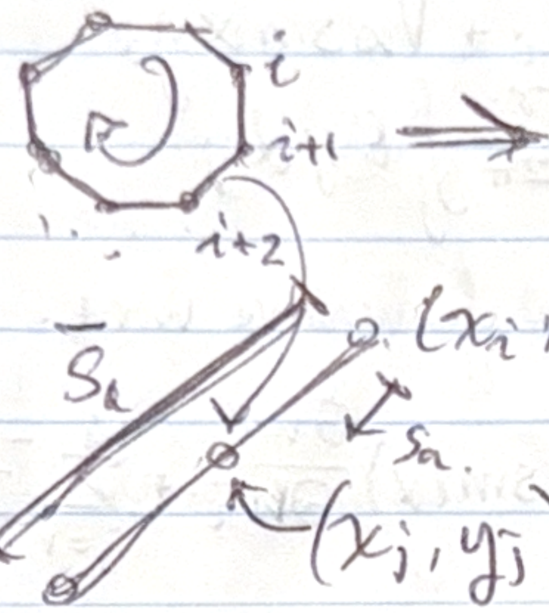
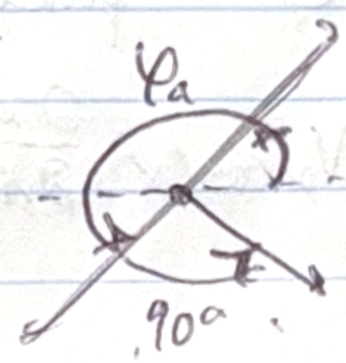
* Siddhartha Somadder (post mtg)

→ there may be errors, fixed in typed-up document.

Clockwise direction ordering.



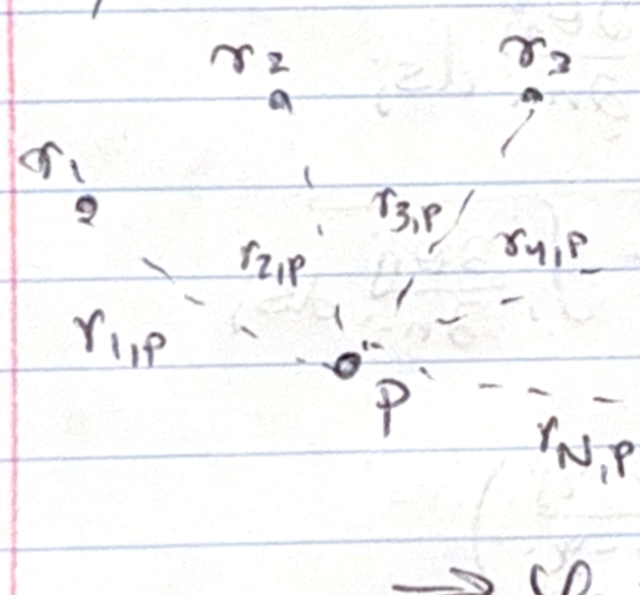
→ control pt: midpoint of panel.
boundary pt: panel nodes.



$\delta_a = \varphi_a + 90°$.
for $\alpha = 0°$;
$\beta_a = \delta_a$
otherwise,
$\beta = \delta_a - \alpha$.

$\varphi_a = \text{atan2}(dy, dx)$
$\bar{s}_a = \|(dx, dy)\|_2$.

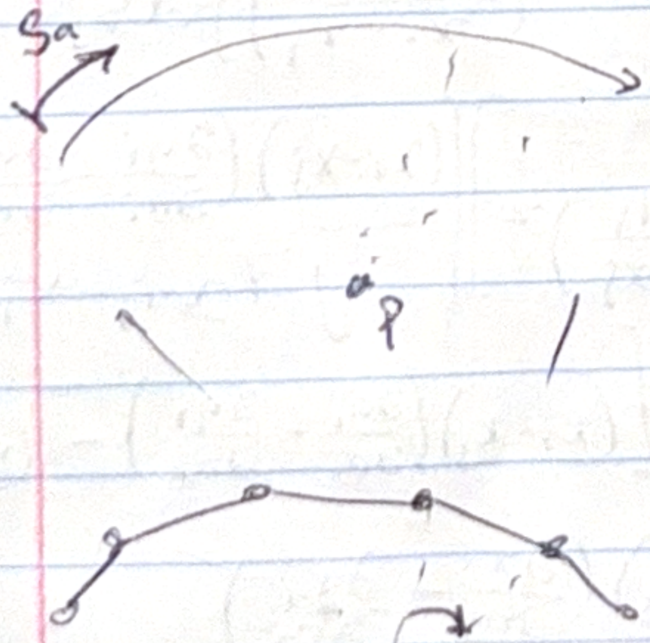
$$\phi = V_\infty \cos(\alpha) x + V_\infty \sin(\alpha) y + \sum_{j=1}^{N} \frac{\gamma_j}{2\pi} \int_j \theta_{ij} \, ds_j$$



using flow superposition,
$\varphi_P = \varphi_1 + \varphi_2 + \dots \varphi_N$

recall φ of a point vortex.
is $\varphi = \frac{-\Gamma}{2\pi}\theta$

$$\Rightarrow \varphi_P = \sum_{i=1}^{N} \frac{-\gamma_i}{2\pi} \theta_{iP}$$



infinitely many $\Rightarrow \varphi_P = \int_a^b \frac{-\gamma(s)}{2\pi} \theta \, ds$.
$\gamma \to \gamma(s)$

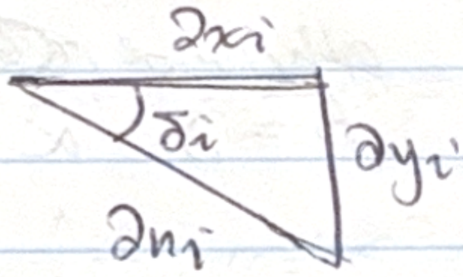
$$\Rightarrow \sum_{i=1}^{N} \frac{-\gamma_i}{2\pi} \int_a^b \theta_{iP} \, ds$$

with uniform contribution:
$$\varphi_P = V_\infty \cos\alpha x + V_\infty \sin\alpha y + \sum_{j=1}^{N} \frac{\gamma_j}{2\pi} \int_j \theta_{ij} \, ds_j$$

Arbitrary panel: $\varphi_i = V_\infty \cos\alpha\, x_i + V_\infty \sin\alpha\, y_i + \sum_{j=1}^{N} \frac{-\gamma_j}{2\pi} \int_j \theta_{ij}\, ds_j$

normal velocity @ ith panel: take derivative w.r.t. n_i

$* V_{n,i} = \frac{\partial \varphi_i}{\partial n_i} = V_\infty \cos(\alpha) \frac{\partial x_i}{\partial n_i} + V_\infty \sin(\alpha) \frac{\partial y_i}{\partial n_i} + \sum_{j=1}^{N} \frac{-\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial n_i} ds_j$



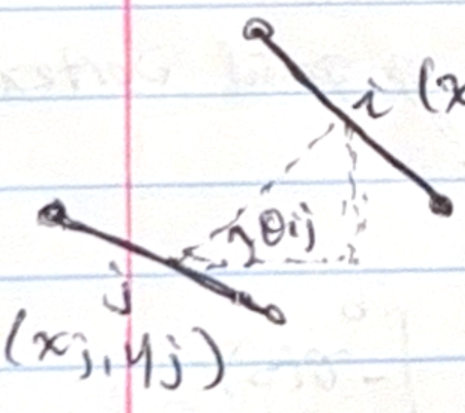
$\rightarrow \cos(\delta_i) = \frac{\partial x_i}{\partial n_i}, \quad \sin(\delta_i) = \frac{\partial y_i}{\partial n_i}.$

$\Rightarrow V_{n,i} = V_\infty (\cos\alpha \cos(\delta_i) + \sin(\alpha)\sin(\delta_i)) + \text{"}\ldots\text{"}$
$= V_\infty \cos(\delta_i - \alpha) + \text{"}\ldots\text{"} = V_\infty \cos\beta_i + \text{"}\ldots\text{"}.$

$* V_{n,i} = V_\infty \cos(\beta_i) - \sum_{j=1}^{N} \frac{\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial n_i} ds_j$

---

we want an expression for $K_{ij} = \int_j \frac{\partial \theta_{ij}}{\partial n_i} ds_j$



$* \theta_{ij} = \tan^{-1}\left(\frac{y_i - y_j}{x_i - x_j}\right)$

$\Rightarrow K_{ij} = \int_j \frac{\partial}{\partial n_i} \left(\tan^{-1}\left(\frac{y_i - y_j}{x_i - x_j}\right)\right) ds_j$

$* \frac{\partial}{\partial n_i} \tan^{-1}\left(\frac{y_i - y_j}{x_i - x_j}\right) = \frac{1}{1 + \left(\frac{y_i - y_j}{x_i - x_j}\right)^2} \left[\frac{(x_i - x_j)\left(\frac{\partial y_i}{\partial n_i} - \frac{\partial y_j}{\partial n_i}\right) + \ldots}{(x_i - x_j)^2}\right]$

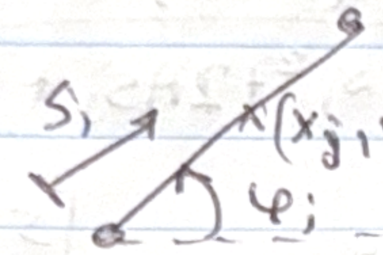
$* \frac{1}{1 + \left(\frac{y_i - y_j}{x_i - x_j}\right)^2} = \frac{(x_i - x_j)^2}{(x_i - x_j)^2 + (y_i - y_j)^2}$

$= \frac{(x_i - x_j)\left(\frac{\partial y_i}{\partial n_i} - \frac{\partial y_j}{\partial n_i}\right) - (y_i - y_j)\left(\frac{\partial x_i}{\partial n_i} - \frac{\partial x_j}{\partial n_i}\right)}{(x_i - x_j)^2 + (y_i - y_j)^2}$

$* \cos(\delta_i) = \frac{\partial x_i}{\partial n_i}, \quad \sin(\delta_i) = \frac{\partial y_i}{\partial n_i}$

$$K_{ij} = \int_j \frac{(x_i - x_j)\sin(\delta_i) - (y_i - y_j)\cos(\delta_i)}{(x_i - x_j)^2 + (y_i - y_j)^2} ds_j$$

Substitute jth panel coordinates in terms of panel arclength and boundary point.



$$x_j = X_j + s_j\cos(\varphi_j)$$
$$y_j = Y_j + s_j\sin(\varphi_j)$$

note $\cos(\delta) = \cos(\varphi + 90°) = -\sin\varphi$.
$\sin(\delta) = \sin(\varphi + 90°) = \cos\varphi$

let $K_{ij} = \int_j \frac{f(s_j)}{g(s_j)} ds_j$, $f = (x_i - x_j)\sin(\delta_i) - (y_i - y_j)\cos(\delta_i)$

$* f = (x_i - (X_j + s_j\cos(\varphi_j))\cos\varphi_i + (y_i - (Y_j + s_j\sin(\varphi_j))\sin\varphi_i$

$= x_i\cos\varphi_i - X_j\cos\varphi_i - s_j\cos(\varphi_i)\cos(\varphi_j) + y_i\sin(\varphi_i)$
$- Y_j\sin\varphi_i - s_j\sin(\varphi_i)\sin(\varphi_j)$

$* \cos\varphi_i(x_i - X_j) + (y_i - Y_j)\sin\varphi_i - s_j(\cos(\varphi_i)\cos(\varphi_j)$
$+\sin(\varphi_i)\sin(\varphi_j))$

$\Rightarrow (x_i - X_j)\cos\varphi_i + (y_i - Y_j)\sin(\varphi_i) - s_j\cos(\varphi_i - \varphi_i)$.

* Numerator: $Cs + D$

$C = -\cos(\varphi_i - \varphi_j)$, $D = (x_i - X_j)\cos\varphi_i + (y_i - Y_j)\sin(\varphi_i)$

Denominator $g = (x_i - x_j)^2 + (y_i - y_j)^2$.

$= (x_i - (X_j + s_j\cos\varphi_j))^2 + (y_i - (Y_j + s_j\sin\varphi_j))^2$.

$(a-b-c)^2 = a^2 + b^2 + c^2 - 2ab - 2ac + 2bc$

$x_i^2 + X_j^2 + s_j^2\cos\varphi_j^2 - 2x_iX_j - 2s_jx_i\cos\varphi_j + 2X_js_j\cos\varphi_j$
$+ y_i^2 + Y_j^2 + s_j^2\sin\varphi_j^2 - 2y_iY_j - 2s_jy_j\sin\varphi_j + 2Y_js_j\sin\varphi_j$

$= s_j^2 + 2s_j((-x_i + X_j)\cos\varphi_j + (-y_i + Y_j)\sin\varphi_j) + (x_i - X_j)^2 + (y_i - Y_j)^2$.

Denominator : $s_j^2 + 2As_j + B$

where $A = -(x_i - x_j)\cos\varphi_j - (y_i - y_j)\sin\varphi_j$

$B = (x_i - x_j)^2 + (y_i - y_j)^2$

$$* \quad K_{ij} = \int_0^S \frac{Cs_j + D}{s_j^2 + 2As_j + B}\, ds_j = \int_0^S \frac{C(s + A) + (D - AC)}{s_j^2 + 2As_j + B}\, ds_j$$

$$\Rightarrow K_{ij} = \frac{C}{2}\int_0^S \frac{2s_j + 2A}{s_j^2 + 2As_j + B}\, ds_j + \int_0^S \frac{D - AC}{s_j^2 + 2As_j + B}\, ds_j$$

let $u = s_j^2 + 2As_j + B$, $du = (2s_j + 2A)\, ds_j$.

and note $s_j^2 + 2As_j + B = s_j^2 + 2As_j + A^2 + B - A^2$

$= (s_j + A)^2 + (B - A^2)$.

<u>first term</u>: $K_{ij,1} = \frac{C}{2}\int_0^S \frac{1}{u}\, du = \frac{C}{2}\ln(u)\Big|_0^S$

$= \frac{C}{2}\ln(S^2 + 2AS + B) - \frac{C}{2}\ln(B)$

$= \frac{C}{2}\ln\left(\frac{S^2 + 2AS + B}{B}\right)$

<u>second term</u>: $K_{ij,2} = \int_0^S \frac{D - AC}{(s_j + A)^2 + (B - A^2)}\, ds_j$

$= D - AC \cdot \int_0^S \frac{ds_j}{(s_j + A)^2 + (B - A^2)}$ let $B - A^2 = E^2$.

$= D - AC \cdot \frac{1}{\sqrt{B - A^2}} \cdot \arctan\left(\frac{s_j + A}{\sqrt{B - A^2}}\right)\Big|_0^S$

$= \frac{D - AC}{\sqrt{B - A^2}}\left[\arctan\left(\frac{S + A}{\sqrt{B - A^2}}\right) - \arctan\left(\frac{A}{\sqrt{B - A^2}}\right)\right]$.

$$* K_{ij} = \frac{C}{2}\ln\left(\frac{S^2 + 2AS + B}{B}\right) + \frac{D - AC}{E}\left[\arctan\left(\frac{S + A}{E}\right) - \arctan\left(\frac{A}{E}\right)\right]$$

$C = -\cos(\varphi_i - \varphi_j)$, $D = (x_i - x_j)\cos\varphi_i + (y_i - y_j)\sin\varphi_i$

$A = -(x_i - x_j)\cos\varphi_j - (y_i - y_j)\sin\varphi_j$

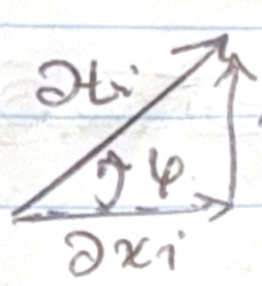
$B = (x_i - x_j)^2 + (y_i - y_j)^2$

$E = (B - A^2)^{1/2}$

$$V_{t,i} = \frac{\partial P_i}{\partial t_i} = V_\infty \sin(\beta_i) - \sum_{j=1}^{N} \frac{\gamma_i}{2\pi} L_{ij}$$

where $L_{ij} = \int_j \frac{\partial \theta_{ij}}{\partial t_i} ds_j$

where



$$\begin{cases} \frac{\partial x_i}{\partial t_i} = \cos(\varphi) \\ \frac{\partial y_i}{\partial t_i} = \sin(\varphi) \end{cases}$$

Similarly

$$L_{ij} = \frac{C}{2} \ln\left(\frac{S^2 + 2AS + B}{B}\right) + \frac{D - AC}{E}\left[\arctan\left(\frac{S+A}{E}\right) - \arctan\left(\frac{A}{E}\right)\right]$$

$$A = -(x_i - x_j)\cos(\varphi_j) - (y_i - y_j)\sin(\varphi_j)$$

$$B = (x_i - x_j)^2 + (y_i - y_j)^2$$

$$C = \sin(\varphi_i - \varphi_j)$$

$$D = (x_i - x_j)\sin(\varphi_i) + (y_i - y_j)\cos(\varphi_i)$$

$$E = \sqrt{B - A^2}$$

$$u_1 = \frac{u_1 + u_2}{2} + \frac{u_2 - u_1}{2}$$

$$= \bar{u} + \frac{\gamma}{2}$$

$$+ \frac{\gamma_i}{2}$$

Using zero-normal velocity condition

$$V_{n,i} = V_\infty \cos(\beta_i) - \sum_{j \neq i}^{N} \frac{\gamma_i}{2\pi} K_{ij} = 0.$$

$$\Rightarrow \sum_{j=1}^{N} \frac{\gamma_j}{2\pi} K_{ij} = V_\infty \cos(\beta_i)$$

$$L' = \rho_\infty V_\infty \Gamma$$

$$C_l q_\infty c = \rho_\infty V_\infty \Gamma.$$

$$C_l \Rightarrow \frac{2\pi}{V_\infty c} = 2\pi$$

$$\Rightarrow$$ [illegible]

$$\begin{pmatrix} 0 & \gamma_2 K_{12} & \gamma_3 K_{13} & \cdots & \gamma_N K_{1N} \\ \gamma_1 K_{21} & 0 & \gamma_3 K_{23} & \cdots & \gamma_N K_{2N} \\ \vdots & & & & \\ \gamma_1 K_{N1} & \gamma_2 K_{N2} & \gamma_3 K_{N3} & \cdots & 0 \end{pmatrix} = \begin{pmatrix} V_\infty \cos(\beta_1) \\ V_\infty \cos(\beta_2) \\ \vdots \\ V_\infty \cos(\beta_N) \end{pmatrix}$$